LAB 7: Fuzzing for Input Validation Bugs

Deliverable 1: Execution & Test Verification Report

Repository: L6_L7_06_event_ticket_tier-main | Framework: Hypothesis + Pytest

Step 2: Initial Execution of Buggy processor.py

Running original unhardened script with nominal inputs (EARLY-20, VIP@4, Alice,25,A12):

```
PS C:\Users\sheno\Downloads\L6_L7_06_event_ticket_tier-main\Lab 7> python processor.py
==== Running decode_promo_voucher ====
Enter voucher (e.g. EARLY-20): EARLY-20
Voucher: Valid EARLY: 20% OFF

==== Running calculate_seat_tier_price ====
Enter tier spec (e.g. VIP@4): VIP@4
Seat Price: $ 158.0

==== Running parse_attendee_badge ====
Enter badge CSV (Name,Age,Seat): Alice,25,A12
Badge: Alice [Adult] - Seat A12
PS C:\Users\sheno\Downloads\L6_L7_06_event_ticket_tier-main\Lab 7>
```

Step 4: Hypothesis Fuzz Testing (Exposing Crashes & Failures)

Running pytest test_processor.py where Hypothesis generates malformed strings and NoneType:

```
=========================================== short test summary info ============================================
FAILED test_processor.py::test_decode_promo_voucher_no_crash - ExceptionGroup: Hypothesis found 3 disti
nct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_seat_tier_price_safe - ExceptionGroup: Hypothesis found 3 dist
inct failures. (3 sub-exceptions)
FAILED test_processor.py::test_parse_attendee_badge_safe - ExceptionGroup: Hypothesis found 3 distinct
failures. (3 sub-exceptions)
============================================= 3 failed in 1.55s ================================================
PS C:\Users\sheno\Downloads\L6_L7_06_event_ticket_tier-main\Lab 7>
```

Step 5: Hardened processor.py Execution

Running the defensively hardened application with sample inputs:

```
PS C:\Users\sheno\Downloads\L6_L7_06_event_ticket_tier-main\Lab 7> python processor.py
==== Running decode_promo_voucher ====
Enter voucher (e.g. EARLY-20): Early-20
Voucher: Valid EARLY: 20% OFF

==== Running calculate_seat_tier_price ====
Enter tier spec (e.g. VIP@4): VIP@4
Seat Price: $ 158.0

==== Running parse_attendee_badge ====
Enter badge CSV (Name,Age,Seat): Abcd,27,9
Badge: Abcd [Adult] - Seat 9
```

Step 6: Hypothesis Fuzz Testing Post-Fix (All Tests Passing)

Re-running pytest test_processor.py with 100+ random fuzz inputs per test:

